

LIGHT TOUCH

James Jeans' Almost-Atomic Theory

Greg Gbur

If history had taken a slightly different turn, we might be talking about the Jeans model of the atom.



James Jeans

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This year marks the 100th anniversary of one of the most important physical theories of all time: Niels Bohr's quantized model of the atom. Building on Ernest Rutherford's observation that the atom consists of a nucleus surrounded by electrically bound electrons, Bohr proposed that the electrons could orbit stably, without radiation, in states of quantized angular momentum.

Using simple dimensional analysis, Jeans demonstrated that no electromagnetic model can possibly produce the discrete atomic frequencies observed in experiments.

With this hypothesis, Bohr was able to successfully reproduce the formula describing the absorption and radiation spectrum of hydrogen, solving what had been a major puzzle in physics for decades. Though Bohr's orbiting electrons were quickly supplanted by a wave picture of electron orbitals, his model was a necessary stepping stone to a quantum theory of matter and quantum mechanics.

Other scientists were well positioned to have similar insights. One of these was English physicist and philosopher James Hopwood Jeans, who

observed in 1906 that existing physics simply could not account for the radiation properties of atoms. If he had carried out his calculations a little differently, Jeans might have uncovered key secrets of the atom years before Bohr.

The early 20th century was a time of great activity—and confusion—in the study of atomic structure. Very little was known about the constitution of atoms, though scientists had a few tantalizing hints: The presence of electrons had been confirmed; different atomic species could be classified; and some heavy atoms produced radioactivity. The most quantitative information, however, came from the discrete

radiation spectrum of hydrogen atoms described using the Rydberg formula, which was empirically derived in 1880:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right),$$

with λ the wavelength of radiation, R the (measured) Rydberg constant, and n_1 and n_2 positive integers.

All previous attempts to describe the atom relied on classical mechanics and electromagnetism. (See sidebar.) Then, in 1906, James Jeans published “On

the constitution of the atom” in *Philosophical Magazine*. Using simple dimensional analysis, Jeans demonstrated that no electromagnetic model can possibly produce the discrete atomic frequencies observed in experiments. As Jeans himself put it,

“If this were so, these frequencies would depend only on the constituents of the atom and not on the actual type of motion taking place in the atom. Thus if we regard the atom as made up of point-charges influencing one another according to the usual electro-dynamical laws, the frequencies could depend only on the number, masses, and charges of the point-charges and on the aether-constant V ... it is impossible, by combining these quantities in any way, to obtain a quantity of the physical dimensions of frequency.”

We may illustrate Jeans’ point with some dimensional analysis. A classical atomic model can only depend on the mass m of the electron, with dimensions M , the charge e of the electron, with dimensions Q , and the “aether-constant” (which we would today call the permittivity, ϵ_0), with dimensions $Q^2 T^2 / ML^3$. Here M represents a mass, Q a charge, T time, and L length. No products of m , e and ϵ_0 or their powers can result in a quantity with dimensions $1/T$ of frequency. This suggested to Jeans that new physics were required to explain the radiation properties of atoms.

His solution was to introduce an electron of finite radius r , which would add a dimension of length L to the system and allow one to combine all quantities into a frequency. This approach did not lead anywhere, however, and the work seems to have been quickly forgotten.

But what could have happened if Jeans had introduced a different parameter into his analysis? By 1906, the relationship between Planck’s constant, h , and a quantized theory of radiation had been

well established, both through Planck’s 1900 theory of blackbody radiation and Einstein’s 1905 theory of the photoelectric effect. If we used Planck’s constant in Jeans’ dimensional analysis, instead of an electron radius ...

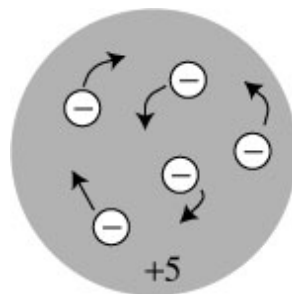
Planck’s constant has dimensions ML^2/T ; we look for solutions of the dimensional equation

$$\epsilon_0^\alpha h^\beta e^\gamma m^\delta = 1/T,$$

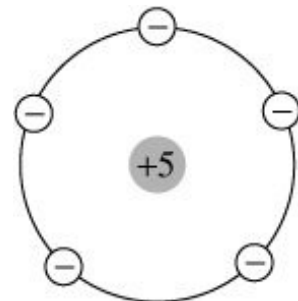
with $\alpha, \beta, \gamma, \delta$ constants to be determined. We quickly find that there is

EARLY ATOMIC MODELS

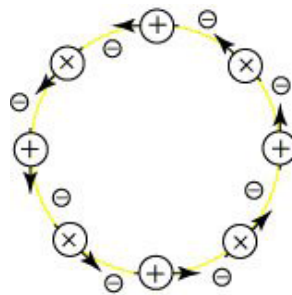
In the early 1900s, some of the greatest minds of the era concocted models to explain the observed properties of atoms, including:



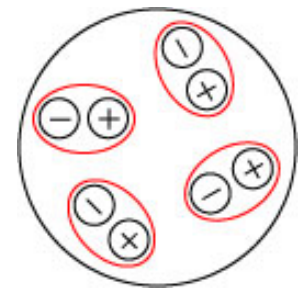
J.J. Thomson’s
“plum pudding” model



H. Nagaoka’s
“Saturnian” model



P. Lenard’s
“dynamid” model



J. Stark’s
“Archion” model

To learn more, visit the author’s blog: <http://skullsinthestars.com/2008/05/27/the-gallery-of-failed-atomic-models-1903-1913/>.

a unique frequency associated with this analysis of the form

$$\nu = \frac{e^4 m}{\epsilon_0^2 h^3},$$

or an equivalent wavelength given by

$$\frac{1}{\lambda} = \frac{e^4 m}{\epsilon_0^2 h^3 c},$$

with c the vacuum speed of light. Let us compare this quantity to the Rydberg constant R as derived from a solution of the Schrödinger equation for the hydrogen atom, namely

$$R = \frac{e^4 m}{8 \epsilon_0^2 h^3 c}.$$

By a simple dimensional analysis, we have derived the form of the Rydberg constant, missing only the numerical factor of $1/8$! If Jeans had used Planck's constant, the relationship to the experimental observations

would have been obvious. This is not to say that Jeans or others would have necessarily beaten Bohr to an atomic model, but the early history of quantum mechanics might have followed a somewhat different path.

Jeans' work was not the only close call in early atomic theory. In 1901, Jean-Baptiste Perrin introduced a "nucleo-planetary" model, pre-dating Rutherford's discovery of the nucleus by nearly a decade. Perrin did not pursue the concept, however, as he could not explain the lack of radiation that should be expected from orbiting electrons. He rightly gave Rutherford full credit; Rutherford, in turn, graciously acknowledged Perrin as the first to introduce the model.

Such "what ifs" illustrate the thin line between scientific discovery and failure in the history of physics. Only

Bohr was bold enough to introduce an entirely new set of laws to break the stalemate in atomic theory—and sometimes, in the end, that makes all the difference. **OPEN**

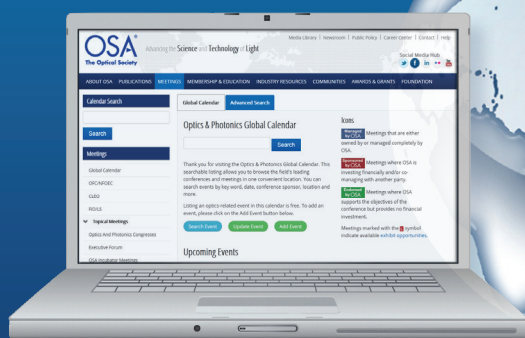
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To learn more ...

- ▶ J.-B. Perrin. Discontinuous Structure of Matter, 1926 Nobel Lecture www.nobelprize.org/nobel_prizes/physics/laureates/1926/perrin-lecture.html.
- ▶ A. Keller. *The Infancy of Atomic Physics*, Clarendon Press, Oxford University Press, United Kingdom, 1983.
- ▶ N. Bohr. *On the Quantum-Theory of Line-Spectra*, Dover Publications, N.Y., 2005.
- ▶ J. Perrin. "Les hypothèses moléculaires," *Revue Scientifique*, 4th series, **15**, 449–61, 1901.

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